

# TOPICS IN MATHEMATICAL SCIENCE

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### INTRODUCTION TO KOSZUL DUALITY

AARON CHAN

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## Convention

Throughout the course,  $\mathbb{k}$  is a field. All rings are unital and associative. All modules are right modules unless otherwise specified. When we discuss dimensions, we mean dimensions as  $\mathbb{k}$ -vector spaces.

## 1 Rings and modules

### 1.1 Semisimple algebras

**Definition 1.1.** A  $\mathbb{k}$ -algebra is a  $\mathbb{k}$ -vector space with an associative unital multiplication that is *bilinear*, meaning linear in each variable.

A right  $A$ -module is understood to be a  $\mathbb{k}$ -vector space  $M$  with a bilinear action satisfying  $(ma)b = m(ab)$  and  $m1_A = m$  for  $m \in M$  and  $a, b \in A$ .

**Example 1.2.** (1) The matrix ring  $\text{Mat}_n(\mathbb{k})$  is a  $\mathbb{k}$ -algebra. Every row space (the space of  $1 \times n$ -matrices) is a  $\text{Mat}_n(\mathbb{k})$ -module.

(2) Let  $\mathbb{k}A_n$  be the *upper triangular matrix ring*, which is also a  $\mathbb{k}$ -algebra.

$$\mathbb{k}A_n = \begin{pmatrix} \mathbb{k} & \mathbb{k} & \cdots & \mathbb{k} \\ 0 & \mathbb{k} & \cdots & \vdots \\ 0 & 0 & \ddots & \vdots \\ 0 & \cdots & 0 & \mathbb{k} \end{pmatrix}$$

For those familiar with path algebras,  $\mathbb{k}A_n$  is the path algebra of a *linearly oriented  $A_n$  quiver*, which explains the notation. We use matrix calculations, so no knowledge of quivers is required.

**Definition 1.3.** A module  $L$  is *simple* if  $L \neq 0$  and its only submodules are 0 and  $L$ . A module is *semisimple* if it is a direct sum of simple modules. An algebra  $B$  is *semisimple* if it is semisimple as a module  $B_B$  over itself (this does not depend on which side you work on). The *Jacobson radical*  $\text{rad } B$  is the intersection of all maximal right ideals of  $B$ .

We use the following standard facts without proof. Recall that a *division  $\mathbb{k}$ -algebra* is a non-zero  $\mathbb{k}$ -algebra whose non-zero elements are invertible.

**Theorem 1.4 (Artin–Wedderburn).** For a non-zero finite-dimensional  $\mathbb{k}$ -algebra  $B$ ,

$$B \text{ is semisimple} \iff \text{rad } B = 0 \iff B \cong \prod_{i=1}^r \text{Mat}_{n_i}(D_i),$$

where  $r, n_i \geq 1$  and each  $D_i$  is a finite-dimensional division  $\mathbb{k}$ -algebra. If  $\mathbb{k}$  is algebraically closed, then  $D_i = \mathbb{k}$  for every  $i$ . For semisimple  $B$ , every finite-dimensional left or right  $B$ -module is semisimple, and every submodule has a complement.

**Example 1.5.** Let  $e_{ij}$  be the unit matrix with 1 at the  $(i, j)$ -entry and zero everywhere else. Then we have  $\text{rad}(\mathbb{k}A_2) = \mathbb{k}e_{12} \neq 0$ , and so  $\mathbb{k}A_2$  is not semisimple.

## 1.2 Graded algebras

**Definition 1.6.** Let  $G$  be a group with identity  $e$ . A  $G$ -graded  $\mathbb{k}$ -algebra is a  $\mathbb{k}$ -algebra  $A$  with  $1_A \in A_e$  and a decomposition

$$A = \bigoplus_{g \in G} A_g \quad \text{such that} \quad A_g A_h \subset A_{gh} \quad \text{for all } g, h \in G.$$

An element of  $A_g$  is *homogeneous of degree  $g$* .

**Convention.** Unless stated otherwise, a graded algebra is  $\mathbb{Z}$ -graded and concentrated in non-negative degrees.

Note that the decomposition is into a direct sum, so every element of a graded algebra  $A$  can always be written as a *finite* sum of homogeneous elements.

**Example 1.7.** For the upper triangular matrix ring  $\mathbb{k}A_n$  of Example 1.2, the decomposition

$$(\mathbb{k}A_n)_d = \text{span}_{\mathbb{k}}\{e_{ij} \mid j - i = d\}$$

makes  $\mathbb{k}A_n$  a graded algebra.

**Example 1.8.** The algebra  $\mathbb{k}A_3$  also has a  $\mathbb{Z}^{\oplus 2}$ -grading with

$$\deg e_{ii} = (0, 0), \quad \deg e_{12} = (1, 0), \quad \deg e_{23} = (0, 1), \quad \deg e_{13} = (1, 1).$$

The homomorphism  $\mathbb{Z}^{\oplus 2} \rightarrow \mathbb{Z}$ ,  $(p, q) \mapsto p + q$ , gives the grading of Example 1.7. Alternatively, setting  $\deg e_{ij} = 2(j - i)$  gives a  $\mathbb{Z}$ -grading with  $A_1 = 0$  but  $A_2 \neq 0$ .

**Definition 1.9.** A *homomorphism of graded algebras*  $f : A \rightarrow B$  is a  $\mathbb{k}$ -algebra homomorphism with  $f(A_i) \subset B_i$  for all  $i \in \mathbb{Z}$ .

**Definition 1.10.** A graded algebra  $A$  is *locally finite* if  $\dim_{\mathbb{k}} A_i < \infty$  for every  $i \in \mathbb{Z}$ .

## 1.3 Graded modules

**Definition 1.11.** Let  $A = \bigoplus_{g \in G} A_g$  be a  $G$ -graded algebra. A  $G$ -graded right  $A$ -module is a right  $A$ -module  $M$  with a direct sum decomposition

$$M = \bigoplus_{g \in G} M_g \quad \text{such that} \quad M_g A_h \subset M_{gh} \quad \text{for all } g, h \in G.$$

An element of  $M_g$  is called *homogeneous of degree  $g$* .

A  $G$ -graded  $A$ -module  $M = \bigoplus_{g \in G} M_g$  is *locally finite-dimensional* if  $\dim_{\mathbb{k}} M_g < \infty$  for all  $g \in G$ .

Unless specified otherwise, graded modules are  $\mathbb{Z}$ -graded; their degrees need not be non-negative. Write  $\text{Mod } A$  for the category of right  $A$ -modules,  $\text{GrMod } A$  for graded right  $A$ -modules with degree 0 homomorphisms, and  $\text{grmod } A$  for its full subcategory of locally finite-dimensional modules.

The *opposite algebra*  $A^{\text{op}}$  has the same vector space as  $A$ , with multiplication  $a^{\text{op}} b^{\text{op}} = (ba)^{\text{op}}$ . A left  $A$ -module is a right  $A^{\text{op}}$ -module by  $m \cdot a^{\text{op}} = am$ . For a  $\mathbb{Z}$ -graded algebra,  $(A^{\text{op}})_i = A_i$ .

**Definition 1.12.** An *A-bimodule* is a vector space  $M$  with  $\mathbb{k}$ -bilinear left and right  $A$ -module actions satisfying  $(am)b = a(mb)$  for all  $a, b \in A$  and  $m \in M$ . It is graded if  $M = \bigoplus_{j \in \mathbb{Z}} M_j$  and  $A_i M_j A_k \subset M_{i+j+k}$  for all  $i, j, k \in \mathbb{Z}$ .

The algebra  $A$  is a graded  $A$ -bimodule by multiplication, and each  $A_i$  is an  $A_0$ -bimodule. The left and right scalar actions agree:  $(c1_A)m = cm = m(c1_A)$  for  $c \in \mathbb{k}$ .

**Example 1.13.** The decomposition  $A = \bigoplus_{i \in \mathbb{Z}} A_i$  makes the right regular module  $A_A$  a graded module.

**Definition 1.14.** Let  $M$  be a graded  $A$ -module and let  $d \in \mathbb{Z}$ . The *shift*  $M(d)$  is the graded  $A$ -module whose underlying  $A$ -module is  $M$ , but whose grading is

$$M(d)_j = M_{j+d}.$$

By Definition 1.14,  $1 \in A_0 = A(-d)_d$ , so  $A(-d)$  has a generator in degree  $d$ .

**Example 1.15.** For  $A = \mathbb{k}A_3$  with the grading of Example 1.7, the module  $P_1 := e_{11}A$  consists of matrices supported on the first row. Its basis  $e_{11}, e_{12}, e_{13}$  has degrees 0, 1, 2, respectively; in  $P_1(-2)$ , the same basis has degrees 2, 3, 4.

**Definition 1.16.** Let  $M, N$  be graded  $A$ -modules. A homomorphism  $f : M \rightarrow N$  is *homogeneous of degree  $d$*  if  $f(M_j) \subset N_{j+d}$  for all  $j \in \mathbb{Z}$ . A *homomorphism of graded modules* means a homogeneous  $A$ -module homomorphism of degree 0, also said to *preserve degrees*.

**Notation.** For graded right  $A$ -modules  $M, N$ ,  $\text{Hom}_A(M, N)$  denotes all right  $A$ -module homomorphisms, without a grading condition. The space of homomorphisms preserving degrees is

$$\text{Hom}_A^{\mathbb{Z}}(M, N) = \{f \in \text{Hom}_A(M, N) \mid f(M_j) \subset N_j \text{ for all } j \in \mathbb{Z}\}. \quad (1.1)$$

Thus  $\text{Hom}_A^{\mathbb{Z}}(M, N(d))$  consists of homomorphisms  $M \rightarrow N$  of degree  $d$ , since  $N(d)_j = N_{j+d}$ . For left modules we use  $\text{Hom}_{A^{\text{op}}}$  and  $\text{Hom}_{A^{\text{op}}}^{\mathbb{Z}}$ .

Finite sums of homogeneous homomorphisms give a natural inclusion

$$\bigoplus_{d \in \mathbb{Z}} \text{Hom}_A^{\mathbb{Z}}(M, N(d)) \subset \text{Hom}_A(M, N).$$

Equality holds when  $M$  is finitely generated, but need not hold for arbitrary  $M$ .

**Example 1.17.** For  $A = \mathbb{k}A_3$ , left multiplication by  $e_{12}$  gives

$$\begin{aligned} e_{22}A &\longrightarrow e_{11}A, \\ \begin{pmatrix} 0 & 0 & 0 \\ 0 & u & v \\ 0 & 0 & 0 \end{pmatrix} &\longmapsto \begin{pmatrix} 0 & u & v \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (u, v \in \mathbb{k}). \end{aligned}$$

The  $u$  and  $v$  entries have degrees 0, 1 in the source and 1, 2 in the target. Thus the map has degree 1; after replacing the source by  $e_{22}A(-1)$ , it belongs to  $\text{Hom}_A^{\mathbb{Z}}(e_{22}A(-1), e_{11}A)$ . Right  $A$ -linearity follows from associativity of matrix multiplication.

## 1.4 Homogeneous ideals and quotients

**Definition 1.18.** Let  $A = \bigoplus_{i \in \mathbb{Z}} A_i$  be a graded algebra. A right ideal  $I \subset A$  is *homogeneous* if

$$I = \bigoplus_{i \in \mathbb{Z}} (I \cap A_i).$$

Equivalently, whenever  $a \in I$  and  $a = a_0 + \cdots + a_N$  is its decomposition into homogeneous components, each  $a_i$  also belongs to  $I$ .

For quotient algebras we need two-sided ideals. If  $I$  is a homogeneous two-sided ideal, then

$$A/I = \bigoplus_{i \in \mathbb{Z}} A_i / (I \cap A_i)$$

is again a graded algebra.

**Example 1.19.** In  $A = \mathbb{k}A_3$  with the grading of Example 1.7, the two-sided ideal  $I = (e_{13}) = \mathbb{k}e_{13}$  is homogeneous. Writing  $\bar{e}_{ij}$  for the image of  $e_{ij}$ ,

$$(A/I)_0 \cong \mathbb{k}^{\oplus 3}, \quad (A/I)_1 = \mathbb{k}\bar{e}_{12} \oplus \mathbb{k}\bar{e}_{23}, \quad (A/I)_d = 0 \quad (d \geq 2).$$

In particular,  $\bar{e}_{12}\bar{e}_{23} = 0$ .

**Example 1.20.** In  $A = \mathbb{k}A_2$ , the subspace  $I = \mathbb{k}(e_{22} + e_{12})$  is a right ideal:  $Ie_{11} = Ie_{12} = 0$  and  $Ie_{22} = I$ . However,  $e_{22}, e_{12} \notin I$ , so  $I$  is not homogeneous. The right module  $A/I$  therefore does not inherit the quotient grading.

**Definition 1.21.** Let  $A$  be a graded algebra and set

$$A_{>0} := \bigoplus_{i>0} A_i.$$

The subspace  $A_{>0}$  is a homogeneous two-sided ideal, also called the [augmentation ideal](#). We use the term [positive degree ideal](#).

The quotient map, also called the augmentation map,

$$\pi : A \longrightarrow A/A_{>0} \cong A_0, \quad \sum_{i \in \mathbb{Z}} a_i \mapsto a_0,$$

makes  $A_0$  a graded  $A$ -bimodule concentrated in degree 0, with  $a \cdot s \cdot b = \pi(a)s\pi(b)$  for  $a, b \in A$  and  $s \in A_0$ .

## 1.5 Tensor products of vector spaces

**Definition 1.22.** For vector spaces  $U, V$ , the [tensor product](#)  $U \otimes_{\mathbb{k}} V$  is equipped with a bilinear map  $(u, v) \mapsto u \otimes v$  such that every bilinear map  $b : U \times V \rightarrow W$  has a unique linear factorisation:

$$\begin{array}{ccc} U \times V & \xrightarrow{(u,v) \mapsto u \otimes v} & U \otimes_{\mathbb{k}} V \\ & \searrow b & \downarrow \exists! \tilde{b} \\ & & W. \end{array}$$

If  $(u_i)_{i=1}^r$  and  $(v_j)_{j=1}^s$  are bases, then  $(u_i \otimes v_j)_{i,j}$  is a basis of  $U \otimes_{\mathbb{k}} V$ ; hence  $\dim_{\mathbb{k}}(U \otimes_{\mathbb{k}} V) = rs$ . We use the canonical identifications  $(U \otimes_{\mathbb{k}} V) \otimes_{\mathbb{k}} W \cong U \otimes_{\mathbb{k}} (V \otimes_{\mathbb{k}} W)$  and  $\mathbb{k} \otimes_{\mathbb{k}} U \cong U$ , and write  $U^{\otimes 0} = \mathbb{k}$ .

## 1.6 Graded $\mathbb{k}$ -linear duals

A linear functional on a graded vector space need not have a degree: it may be non-zero on infinitely many homogeneous components. To obtain a graded vector space, we retain only finite sums of homogeneous functionals.

**Definition 1.23.** For a vector space  $V$ , its  $\mathbb{k}$ -linear dual is  $D(V) := \text{Hom}_{\mathbb{k}}(V, \mathbb{k})$ . For a graded vector space  $V = \bigoplus_{i \in \mathbb{Z}} V_i$ , its *graded  $\mathbb{k}$ -linear dual* is the graded vector space given by

$$\mathbb{D}(V) = \bigoplus_{i \in \mathbb{Z}} \mathbb{D}(V)_i \quad \text{with} \quad \mathbb{D}(V)_i = D(V_{-i}).$$

For general infinite-dimensional  $V$ , we have  $\mathbb{D}(V) \subset D(V)$ , whereas  $D(V) \cong \prod_{i \in \mathbb{Z}} D(V_i)$ . These spaces coincide when  $V$  is finite-dimensional.

Linear duality reverses the side of a module action. For a right  $A$ -module  $M$  and a left  $A$ -module  $N$ , define

$$(af)(m) := f(ma), \quad (ga)(n) := g(an) \quad (a \in A, f \in D(M), g \in D(N)). \quad (1.2)$$

Here  $m \in M$  and  $n \in N$ ; the first formula makes  $D(M)$  a left module and the second makes  $D(N)$  a right module. For graded modules, the same formulas preserve the graded  $\mathbb{k}$ -linear dual: if  $a \in A_p$  and  $f \in \mathbb{D}(M)_q$ , then  $af$  vanishes outside  $M_{-p-q}$  and hence has degree  $p + q$ . There are no signs in these actions.

**Example 1.24.** Let  $A = \mathbb{k}A_3$  and  $P_1 = e_{11}A$  as in Example 1.15. The functionals  $\varepsilon_{1j}(e_{1\ell}) = \delta_{j\ell}$  for  $1 \leq j, \ell \leq 3$  form a basis of  $\mathbb{D}(P_1)$ , with

$$\deg \varepsilon_{1j} = 1 - j, \quad e_{12}\varepsilon_{12} = \varepsilon_{11}, \quad e_{23}\varepsilon_{13} = \varepsilon_{12}, \quad e_{jj}\varepsilon_{1j} = \varepsilon_{1j}.$$

The actions follow from (1.2). For comparison with an infinite-dimensional space, take  $V = \bigoplus_{i \geq 0} \mathbb{k}v_i$  with  $\deg v_i = i$ . The functional  $\lambda(v_i) = 1$  for every  $i$  belongs to  $D(V)$  but not to  $\mathbb{D}(V)$ .

**Lemma 1.25.**  $\mathbb{D}$  is a duality between  $\text{grmod } A$  and  $\text{grmod } A^{\text{op}}$ : it reverses arrows and short exact sequences, and evaluation gives a graded module isomorphism  $M \cong \mathbb{D}(\mathbb{D}(M))$  for every locally finite graded module  $M$ .

**Proof** A homomorphism  $u : M \rightarrow N$  of degree 0 induces  $\mathbb{D}(u) : \mathbb{D}(N) \rightarrow \mathbb{D}(M)$  by  $f \mapsto f \circ u$ . Dualising a short exact sequence in each degree is exact, since every linear functional on a subspace extends to the whole vector space. The evaluation map sends  $m$  to  $(f \mapsto f(m))$ . It is an isomorphism in each degree by finite-dimensional linear algebra, and (1.2) makes it a module homomorphism.  $\square$

## 1.7 Left, right, and two-sided duals over $S$

Fix a finite-dimensional semisimple  $\mathbb{k}$ -algebra  $S$ , concentrated in degree 0. Besides the  $\mathbb{k}$ -linear dual, we need duals defined by left  $S$ -linear maps, right  $S$ -linear maps, and maps respecting both actions. Their definitions differ, even though the resulting spaces can be identified after a choice of form.

**Definition 1.26.** For a right  $S$ -module  $M$  and a left  $S$ -module  $N$ , define their  *$S$ -duals* by

$$\begin{aligned} M^* &:= \text{Hom}_S(M, S), & (sf)(m) &:= sf(m), \\ {}^*N &:= \text{Hom}_{S^{\text{op}}}(N, S), & (gs)(n) &:= g(n)s. \end{aligned}$$

Here  $s \in S$ ,  $m \in M$ ,  $n \in N$ ,  $f \in M^*$ , and  $g \in {}^*N$ .

Thus  $M^*$  is a left  $S$ -module and  ${}^*N$  is a right  $S$ -module. The position of the star records the side of the original action:  $f(ms) = f(m)s$  in  $M^*$ , whereas  $g(sn) = sg(n)$  in  ${}^*N$ .

**Definition 1.27.** The *enveloping algebra* of  $S$  is  $S^{\text{en}} := S \otimes_{\mathbb{k}} S^{\text{op}}$ , with multiplication

$$(s \otimes t^{\text{op}})(s' \otimes (t')^{\text{op}}) = ss' \otimes (t't)^{\text{op}} \quad (s, t, s', t' \in S).$$

An  $S$ -bimodule is equivalently a right  $S^{\text{en}}$ -module via

$$m \cdot (s \otimes t^{\text{op}}) = tms \quad (s, t \in S, m \in M). \quad (1.3)$$

For example,  $(m \cdot (s \otimes t^{\text{op}})) \cdot (s' \otimes (t')^{\text{op}}) = t'tmss' = m \cdot (ss' \otimes (t't)^{\text{op}})$ . All three spaces  $D(M)$ ,  $M^*$ , and  ${}^*M$  inherit bimodule structures:

$$(s\lambda t)(m) = \lambda(tms), \quad (sft)(m) = sf(tm), \quad (sgt)(m) = g(ms)t, \quad (1.4)$$

where  $\lambda \in D(M)$ ,  $f \in M^*$ ,  $g \in {}^*M$ ,  $s, t \in S$ , and  $m \in M$ .

**Definition 1.28.** The *two-sided dual* of an  $S$ -bimodule  $M$  is

$$M^{*\text{en}} := \text{Hom}_{S^{\text{en}}}(M, S^{\text{en}}), \quad (sFt)(m) := (s \otimes t^{\text{op}})F(m) \quad (s, t \in S, F \in M^{*\text{en}}, m \in M).$$

In Definition 1.28,  $M$  has the right action (1.3), so  $F(tms) = F(m)(s \otimes t^{\text{op}})$ . The output is a left  $S^{\text{en}}$ -module by multiplication on the values, which gives the displayed bimodule action. The two-sided dual takes values in  $S^{\text{en}}$ , not in  $S$ .

## 1.8 Comparing the duals

An  $S$ -dual consists of  $S$ -valued maps; the  $\mathbb{k}$ -linear dual consists of scalar-valued maps. A suitable linear functional  $S \rightarrow \mathbb{k}$  identifies these constructions.

**Definition 1.29.** A *symmetrising form* on a finite-dimensional  $\mathbb{k}$ -algebra  $S$  is a linear map  $\tau : S \rightarrow \mathbb{k}$  satisfying

$$\tau(st) = \tau(ts) \quad \text{for all } s, t \in S, \quad (\tau(st) = 0 \text{ for all } t \in S) \implies s = 0.$$

An algebra admitting such a form is *symmetric*.

We use the standard theorem that every finite-dimensional semisimple  $\mathbb{k}$ -algebra is symmetric. We do not prove this structure theorem here. The forms needed in our basic examples are explicit.

**Example 1.30.** For  $S = \mathbb{k}^{\oplus r}$ , take  $\tau(a_1, \dots, a_r) = \sum_i a_i$ . For  $S = \text{Mat}_n(\mathbb{k})$ , take  $\tau(a) = \text{Tr}(a)$ . Indeed,  $\text{Tr}(ab) = \text{Tr}(ba)$  and  $\text{Tr}(ae_{ji}) = a_{ij}$ , so the pairing is non-degenerate in every characteristic. More generally, the sum of the matrix traces works on every product  $\prod_{j=1}^r \text{Mat}_{n_j}(\mathbb{k})$  in Theorem 1.4.

**Proposition 1.31.** Choose a symmetrising form  $\tau$  on  $S$ . For every finite-dimensional  $S$ -bimodule  $M$ , there are isomorphisms of  $S$ -bimodules

$$\alpha_M : M^* \longrightarrow D(M), \quad f \longmapsto \tau \circ f, \quad \beta_M : {}^*M \longrightarrow D(M), \quad g \longmapsto \tau \circ g. \quad (1.5)$$

There is also an isomorphism  $\gamma_M : M^{*\text{en}} \rightarrow M^*$  given by

$$\gamma_M(F)(m) = \sum_j a_j \tau(b_j) \quad \text{where} \quad F(m) = \sum_j a_j \otimes b_j^{\text{op}}. \quad (1.6)$$

The maps  $\alpha_M, \beta_M, \gamma_M$  commute with precomposition by bimodule homomorphisms. The maps  $\alpha_M$  and  $\beta_M$  also apply to right and left  $S$ -modules, respectively, with the opposite side on the duals.

**Proof** For  $\lambda \in D(M)$ , non-degeneracy of  $\tau$  determines  $f(m), g(m) \in S$  uniquely by

$$\tau(f(m)s) = \lambda(ms), \quad \tau(sg(m)) = \lambda(sm) \quad (s \in S, m \in M). \quad (1.7)$$

Equation (1.7) gives  $f(ms) = f(m)s$ ,  $g(sm) = sg(m)$ , and  $\alpha_M(f) = \beta_M(g) = \lambda$ . Hence  $\alpha_M, \beta_M$  are bijective. For bimodule compatibility, use (1.4) and

$$\tau(sf(tm)) = \tau(f(tm)s) = \tau(f(tms)), \quad \tau(g(ms)t) = \tau(tg(ms)) = \tau(g(tms)).$$

The form  $\tau_{\text{en}}(a \otimes b^{\text{op}}) = \tau(a)\tau(b)$  is symmetrising on  $S^{\text{en}}$ . The formula (1.6) satisfies

$$\gamma_M(F)(ms) = \gamma_M(F)(m)s, \quad \alpha_M(\gamma_M(F)) = \tau_{\text{en}} \circ F.$$

The argument (1.7), applied to  $S^{\text{en}}$  and  $\tau_{\text{en}}$ , makes  $F \mapsto \tau_{\text{en}} \circ F$  an isomorphism. Thus  $\gamma_M$  is an isomorphism. Composition with  $\tau$  and  $\tau_{\text{en}}$  commutes with precomposition, giving naturality.  $\square$

The identifications (1.5) and (1.6) depend on  $\tau$ . For instance,  $\tau(a_1, \dots, a_r) = \sum_i c_i a_i$  is symmetrising on  $\mathbb{k}^{\oplus r}$  for any  $c_i \neq 0$ . No semisimplicity assumption on  $S^{\text{en}}$  was used.

**Definition 1.32.** For a graded  $S$ -bimodule  $M$ , define its *graded  $S$ -duals* and *graded two-sided dual* by

$$M^{(*)} := \bigoplus_{i \in \mathbb{Z}} (M_{-i})^*, \quad ({}^*)M := \bigoplus_{i \in \mathbb{Z}} {}^*(M_{-i}), \quad M^{(*\text{en})} := \bigoplus_{i \in \mathbb{Z}} (M_{-i})^{*\text{en}},$$

where each indicated summand has degree  $i$ . The first construction also applies to graded right  $S$ -modules, and the second to graded left  $S$ -modules.

Parentheses on the stars distinguish the graded constructions from the full  $S$ -duals. In the notation (1.1),

$$M^{(*)} = \bigoplus_{i \in \mathbb{Z}} \text{Hom}_S^{\mathbb{Z}}(M, S(i)), \quad ({}^*)M = \bigoplus_{i \in \mathbb{Z}} \text{Hom}_{S^{\text{op}}}^{\mathbb{Z}}(M, S(i)).$$

**Corollary 1.33.** For a locally finite graded  $S$ -bimodule  $M$ , a choice of symmetrising form gives natural isomorphisms of graded  $S$ -bimodules

$$M^{(*)} \cong \mathbb{D}(M) \cong ({}^*)M \cong M^{(*\text{en})}.$$

In particular,  $\mathbb{D}(M) \cong M^{(*)}$  for a graded right  $S$ -module, and  $\mathbb{D}(M) \cong ({}^*)M$  for a graded left  $S$ -module.

**Proof** Apply Proposition 1.31 to  $M_{-i}$  and take the direct sum over  $i \in \mathbb{Z}$ . Every comparison preserves degree because  $\tau$  has degree 0.  $\square$

For a graded algebra  $A$  with  $S = A_0$ , Corollary 1.33 transports the  $A$ -action (1.2) to the graded  $S$ -duals of a locally finite graded  $A$ -module:

$$af = \alpha_M^{-1}(a\alpha_M(f)) \quad \text{for right } M, \quad ga = \beta_M^{-1}(\beta_M(g)a) \quad \text{for left } M.$$

Here  $\alpha_M(f) = \tau \circ f$  and  $\beta_M(g) = \tau \circ g$  are the restrictions of (1.5);  $a \in A$ ,  $f \in M^{(*)}$ , and  $g \in ({}^*)M$ . The restrictions to  $S$  agree with Definition 1.26. These remain duals over  $S$ , not duals taking values in  $A$ .

**Example 1.34.** Let  $S = \mathbb{k}^{\oplus r}$  with coordinate idempotents  $e_1, \dots, e_r$  and the form  $\tau(e_i) = 1$ . For an  $S$ -bimodule  $M$ , the decompositions and the reversal of sides are

$$M = \bigoplus_{i,j} e_i M e_j, \quad e_j D(M) e_i \cong D(e_i M e_j) \cong e_j M^* e_i \cong e_j ({}^*M) e_i.$$

For the degree 1 piece  $V = \mathbb{k}e_{12}$  of  $\mathbb{k}A_2$ , we have  $e_1 V e_2 = V$ . Its graded  $\mathbb{k}$ -linear dual is spanned by  $\varepsilon$  of degree  $-1$ , with  $\varepsilon(e_{12}) = 1$  and  $e_2 \varepsilon e_1 = \varepsilon$ . The corresponding elements of  $V^*$  and  ${}^*V$  send  $e_{12}$  to  $e_2$  and  $e_1$ , respectively. Thus equal scalar-valued functionals can correspond to different  $S$ -valued maps.

**Lemma 1.35.** For finite-dimensional right  $S$ -modules  $M$  and left  $S$ -modules  $N$ , evaluation gives canonical isomorphisms  $M \cong {}^*(M^*)$  and  $N \cong ({}^*N)^*$ , sending  $m$  to  $(f \mapsto f(m))$  and  $n$  to  $(g \mapsto g(n))$ , respectively. For bimodules, both are bimodule isomorphisms. No symmetrising form is needed.

**Proof** By Theorem 1.4, every finite-dimensional  $S$ -module is a direct summand of some  $S^{\oplus n}$ . Evaluation is an isomorphism for  $S$ , hence for such summands; (1.4) gives bimodule compatibility.  $\square$

## 2 Lecture 2: Tensor and quadratic algebras

### 2.1 Tensor products over an algebra

The degree zero algebra acts on both sides of the degree 1 piece. We therefore need the tensor product over an algebra, not only the tensor product over  $\mathbb{k}$ .

**Definition 2.1.** Let  $M$  be a right  $S$ -module, let  $N$  be a left  $S$ -module, and let  $W$  be a  $\mathbb{k}$ -vector space. The **tensor product**  $M \otimes_S N$  is the quotient of  $M \otimes_{\mathbb{k}} N$  by the subspace generated by

$$ms \otimes n - m \otimes sn \quad (m \in M, n \in N, s \in S).$$

Equivalently, a bilinear map  $b : M \times N \rightarrow W$  factors uniquely through a linear map  $M \otimes_S N \rightarrow W$  precisely when  $b(ms, n) = b(m, sn)$  for all  $m \in M, s \in S$ , and  $n \in N$ .

The equality  $ms \otimes_S n = m \otimes_S sn$  allows a scalar from  $S$  to pass between adjacent factors. For an  $S$ -bimodule  $V$ , define

$$V^{\otimes_S 0} := S, \quad V^{\otimes_S i} := \underbrace{V \otimes_S \cdots \otimes_S V}_{i \text{ factors}} \quad (i \geq 1).$$

Tensor duality reverses the order of the factors.

**Proposition 2.2.** Let  $S$  be finite-dimensional semisimple and let  $V, W$  be finite-dimensional  $S$ -bimodules. There are natural bimodule isomorphisms

$$\begin{aligned} \theta_D : D(W) \otimes_S V^* &\xrightarrow{\sim} D(V \otimes_S W), \\ \theta_r : W^* \otimes_S V^* &\xrightarrow{\sim} (V \otimes_S W)^*, \\ \theta_\ell : {}^*W \otimes_S {}^*V &\xrightarrow{\sim} {}^*(V \otimes_S W), \\ \theta_{\text{en}} : W^* \otimes_S V^{*\text{en}} &\xrightarrow{\sim} (V \otimes_S W)^{*\text{en}}, \end{aligned} \tag{2.1}$$

defined by

$$\begin{aligned} \theta_D(\lambda \otimes f)(v \otimes w) &= \lambda(f(v)w), \\ \theta_r(g \otimes f)(v \otimes w) &= g(f(v)w), \\ \theta_\ell(g \otimes f)(v \otimes w) &= f(vg(w)), \\ \theta_{\text{en}}(g \otimes F)(v \otimes w) &= \sum_j g(a_j w) \otimes b_j^{\text{op}}, \quad \text{where } F(v) = \sum_j a_j \otimes b_j^{\text{op}}. \end{aligned} \tag{2.2}$$

**Proof** The formulas in (2.2) respect the tensor relations and bimodule actions. The maps  $\theta_D$  and  $\theta_r$  are isomorphisms for  $V = S$  as a right module, hence for finite direct sums and direct summands of  $S$ . For  $\theta_\ell$ , apply the same argument to the left module  $W$ . Choose a symmetrising form  $\tau$  as in Definition 1.29, and use  $\gamma_V, \gamma_{V \otimes_S W}$  from (1.6). Then

$$\gamma_{V \otimes_S W} \circ \theta_{\text{en}} = \theta_r \circ (\text{id}_{W^*} \otimes \gamma_V),$$

since both sides evaluate  $g \otimes F$  at  $v \otimes w$  as  $\sum_j g(a_j w) \tau(b_j)$ . Proposition 1.31 makes both  $\gamma$  maps isomorphisms, so  $\theta_{\text{en}}$  is an isomorphism.  $\square$

For a fixed symmetrising form  $\tau$ , let  $\alpha_M(f) = \tau \circ f$  as in (1.5). Then tensor evaluation satisfies

$$\alpha_{V \otimes_S W} \circ \theta_r = \theta_D \circ (\alpha_W \otimes \text{id}_{V^*}).$$

Both sides send  $g \otimes f$  to  $v \otimes w \mapsto \tau(g(f(v)w))$ .

**Example 2.3.** Let  $S = \mathbb{k}^{\oplus r}$  with coordinate idempotents  $e_1, \dots, e_r$ , and suppose  $V = e_i V e_j$  and  $W = e_j W e_\ell$ . By (1.4),

$$V \otimes_S W = e_i(V \otimes_S W)e_\ell, \quad W^* \otimes_S V^* = e_\ell(W^* \otimes_S V^*)e_i.$$

Thus  $\theta_r$  in (2.1) reverses the two outer idempotents.

## 2.2 Generated algebras

Let  $A$  be graded with  $S = A_0$ . Multiplication makes  $A_1$  an  $S$ -bimodule and gives

$$\mu_i : A_1^{\otimes_S i} \longrightarrow A_i, \quad a_1 \otimes \dots \otimes a_i \longmapsto a_1 \dots a_i \quad (i \geq 1), \quad \mu_0 = \text{id}_S. \quad (2.3)$$

**Definition 2.4.** The algebra  $A$  is *generated in degree 1 over  $S$*  if every  $\mu_i$  in (2.3) is surjective.

Equivalently,  $S$  and  $A_1$  generate  $A$  as an algebra.

**Example 2.5.** The algebra  $\mathbb{k}A_n$  of Example 1.7 is generated in degree 1 over  $S = \bigoplus_i \mathbb{k}e_{ii}$ :

$$e_{ij} = e_{i,i+1}e_{i+1,i+2} \dots e_{j-1,j} \quad (1 \leq i < j \leq n).$$

For the grading  $\deg e_{ij} = 2(j-i)$ , the same algebra has  $A_1 = 0$  and  $A_2 \neq 0$ , so it is not generated in degree 1.

## 2.3 Tensor algebras and the universal property

**Definition 2.6.** Let  $S$  be a  $\mathbb{k}$ -algebra and  $V$  an  $S$ -bimodule. The *tensor algebra of  $V$  over  $S$*  is

$$T_S(V) := \bigoplus_{i \in \mathbb{Z}} T_S(V)_i, \quad \text{where} \quad T_S(V)_i = \begin{cases} V^{\otimes_S i}, & i \geq 1, \\ S, & i = 0, \\ 0, & i < 0, \end{cases}$$

with multiplication given by concatenation of tensors and the given  $S$ -actions.

The left and right  $S$ -actions determine which factors can be concatenated. For  $S = \bigoplus_{i=1}^r \mathbb{k}e_i$  with coordinate idempotents,

$$V = \bigoplus_{i,j} e_i V e_j, \quad (e_i V e_j) \otimes_S (e_\ell V e_m) = 0 \quad \text{if } j \neq \ell. \quad (2.4)$$

Indeed,  $v \otimes w = v e_j \otimes w = v \otimes e_j w = 0$  when  $v \in e_i V e_j$ ,  $w \in e_\ell V e_m$ , and  $j \neq \ell$ .

**Example 2.7.** Let  $S = \mathbb{k}^{\oplus 3}$  with coordinate idempotents  $e_1, e_2, e_3$ , and let  $V = \mathbb{k}a \oplus \mathbb{k}b$  have actions  $e_1 a e_2 = a$  and  $e_2 b e_3 = b$ , with all other idempotent actions zero. By (2.4),

$$V \otimes_S V = \mathbb{k}(a \otimes b), \quad V^{\otimes_S 3} = 0, \quad T_S(V) = S \oplus \mathbb{k}a \oplus \mathbb{k}b \oplus \mathbb{k}ab.$$

Here  $ab = a \otimes b$ . The assignments  $e_i \mapsto e_{ii}$ ,  $a \mapsto e_{12}$ , and  $b \mapsto e_{23}$  give  $T_S(V) \cong \mathbb{k}A_3$ ; the tensor  $ab$  maps to  $e_{13}$ .

**Proposition 2.8 (Universal property of the tensor algebra).** Let  $\iota_S : S \rightarrow T_S(V)$  and  $\iota_V : V \rightarrow T_S(V)$  be the canonical inclusions. Given a unital  $\mathbb{k}$ -algebra homomorphism  $\varphi : S \rightarrow B$  and an  $S$ -bimodule map  $f : V \rightarrow B$ , where  $sbt = \varphi(s)b\varphi(t)$  on  $B$ , there is a unique  $\mathbb{k}$ -algebra homomorphism  $\tilde{f} : T_S(V) \rightarrow B$  making the diagram commute:

$$\begin{array}{ccccc} V & \xrightarrow{\iota_V} & T_S(V) & \xleftarrow{\iota_S} & S \\ & \searrow f & \downarrow \exists! \tilde{f} & \swarrow \varphi & \\ & & B & & \end{array} \quad (2.5)$$

**Proof** Define  $\tilde{f}$  linearly by

$$\tilde{f}(s) = \varphi(s), \quad \tilde{f}(v_1 \otimes_S \cdots \otimes_S v_i) = f(v_1) \cdots f(v_i) \quad (s \in S, i \geq 1). \quad (2.6)$$

The identity  $f(svt) = \varphi(s)f(v)\varphi(t)$  makes  $\tilde{f}$  well-defined on tensor products and compatible with multiplication. Every homomorphism satisfying (2.5) has the values in (2.6), proving uniqueness.  $\square$

**Corollary 2.9.** *If  $A$  is generated in degree 1 over  $S = A_0$ , there is a surjective graded algebra homomorphism  $\mu : T_S(A_1) \twoheadrightarrow A$  whose degree  $i$  component is  $\mu_i$  from (2.3).*

**Proof** Apply Proposition 2.8 to the inclusions  $S \rightarrow A$  and  $A_1 \rightarrow A$ . Surjectivity of each  $\mu_i$  gives surjectivity of  $\mu$ .  $\square$

**Definition 2.10.** *For the map  $\mu$  in Corollary 2.9, the homogeneous two-sided ideal  $I := \text{Ker } \mu$  is the ideal of relations.*

Writing  $V = A_1$ , we obtain  $A \cong T_S(V)/I$  with  $I_0 = I_1 = 0$ . The bimodule  $V$  specifies the generators and their left and right  $S$ -actions;  $I$  specifies the relations between their products.

## 2.4 Quadratic algebras

**Definition 2.11.** *Let  $S$  be finite-dimensional semisimple,  $V$  a finite-dimensional  $S$ -bimodule, and  $R \subset V \otimes_S V$  an  $S$ -sub-bimodule. The algebra  $T_S(V)/(R)$  is a quadratic algebra, where  $(R)$  denotes the two-sided ideal generated by  $R$ .*

**Proposition 2.12.** *A quadratic algebra  $A = T_S(V)/(R)$  has  $A_0 = S$  and  $A_1 = V$ , is locally finite, and is generated in degree 1 over  $S$ .*

**Proof** The ideal  $(R)$  has no components in degrees 0 and 1. Each  $A_i$  is a quotient of the finite-dimensional space  $V^{\otimes_S i}$ , and multiplication maps  $V^{\otimes_S i}$  onto  $A_i$ .  $\square$

**Example 2.13.** *Let  $S = \bigoplus_{i=1}^n \mathbb{k}e_{ii}$  and  $V = \bigoplus_{i=1}^{n-1} \mathbb{k}e_{i,i+1}$  inside  $\mathbb{k}A_n$ . By (2.4), the basis tensors of positive length are*

$$e_{i,i+1} \otimes_S e_{i+1,i+2} \otimes_S \cdots \otimes_S e_{j-1,j} \mapsto e_{ij} \quad (1 \leq i < j \leq n).$$

*Thus multiplication gives  $T_S(V) \cong \mathbb{k}A_n$ , a quadratic algebra with  $R = 0$ .*

**Example 2.14.** *For  $S, V, a, b$  from Example 2.7, the quotient*

$$T_S(V)/(ab) \cong \mathbb{k}A_3/(e_{13})$$

*is quadratic with  $R = \mathbb{k}(a \otimes b)$ . Its basis is  $e_1, e_2, e_3, a, b$ , in degrees 0, 0, 0, 1, 1, as in Example 1.19.*

**Example 2.15.** *Let  $S = \mathbb{k}^{\oplus 2}$  with coordinate idempotents  $e_1, e_2$ , and let  $V = \mathbb{k}a \oplus \mathbb{k}b$  satisfy  $e_1 a e_2 = a$  and  $e_2 b e_1 = b$ , with all other idempotent actions zero. In  $T_S(V)$  only alternating words in  $a, b$  survive. The quadratic algebra  $A = T_S(V)/(ab)$  has*

$$A = S \oplus \mathbb{k}a \oplus \mathbb{k}b \oplus \mathbb{k}ba, \quad \dim_{\mathbb{k}} A_0 = 2, \quad \dim_{\mathbb{k}} A_1 = 2, \quad \dim_{\mathbb{k}} A_2 = 1.$$

*Every alternating word of length at least 3 contains  $ab$ , so all higher components vanish.*

**Example 2.16.** *Let  $B = \mathbb{k}A_4$  and  $A = B/(e_{14})$ , with the grading of Example 1.7. The ideal  $I = (e_{14}) = \mathbb{k}e_{14}$  has  $I_2 = 0$  but  $I_3 \neq 0$ . Since  $e_{14} = e_{12}e_{23}e_{34}$ , the algebra  $A$  is generated in degree 1 with a relation of degree 3 and no relations of degree 2. Hence  $A$  is not quadratic.*

**Example 2.17 (Polynomial and exterior algebras).** For the classical comparison, take  $S = \mathbb{k}$  and  $V = \bigoplus_{i=1}^n \mathbb{k}x_i$ . The [symmetric algebra](#) and [exterior algebra](#) are

$$\begin{aligned} \text{Sym}(V) &= T_{\mathbb{k}}(V)/(x_i x_j - x_j x_i \mid i < j) \cong \mathbb{k}[x_1, \dots, x_n], \\ \bigwedge V &= T_{\mathbb{k}}(V)/(v^2 \mid v \in V). \end{aligned} \tag{2.7}$$

Both are quadratic. The polynomial basis allows repetitions; the exterior basis consists of 1 and  $x_{i_1} \cdots x_{i_d}$  with  $i_1 < \cdots < i_d$ . The exterior relations are equivalently  $x_i^2 = 0$  for  $1 \leq i \leq n$  and  $x_i x_j + x_j x_i = 0$  for  $i < j$ , in every characteristic.

## 2.5 Hilbert series

A Hilbert series records the dimension of each homogeneous component.

**Definition 2.18.** Let  $M = \bigoplus_{j \in \mathbb{Z}} M_j$  be a graded vector space with  $\dim_{\mathbb{k}} M_j < \infty$  for all  $j$  and  $M_j = 0$  for all  $j \ll 0$ . Its [Hilbert series](#) is the formal Laurent series  $h_M(t) := \sum_{j \in \mathbb{Z}} (\dim_{\mathbb{k}} M_j) t^j$ .

For a locally finite graded algebra  $A$ ,  $h_A(t)$  refers to its underlying graded vector space; the series does not record multiplication.

**Lemma 2.19.** Let  $M, N$  be locally finite graded vector spaces concentrated in degrees bounded below.

- (1) A graded isomorphism  $M \cong N$  implies  $h_M(t) = h_N(t)$ .
- (2) An exact sequence  $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$  with degree 0 maps gives  $h_M(t) = h_L(t) + h_N(t)$ .
- (3) For every  $d \in \mathbb{Z}$ ,  $h_{M(d)}(t) = t^{-d} h_M(t)$ .

**Proof** The first two statements follow from dimensions in each degree. For shifts,

$$h_{M(d)}(t) = \sum_j \dim_{\mathbb{k}}(M_{j+d}) t^j = t^{-d} \sum_i \dim_{\mathbb{k}}(M_i) t^i. \quad \square$$

**Example 2.20.** For the upper triangular algebra in Example 1.7, counting matrix entries on each diagonal gives

$$h_{\mathbb{k}A_n}(t) = n + (n-1)t + \cdots + t^{n-1}, \quad h_{\mathbb{k}A_3/(e_{13})}(t) = 3 + 2t.$$

The second equality also follows by subtracting  $h_{(e_{13})}(t) = t^2$  from  $h_{\mathbb{k}A_3}(t)$ , as in Lemma 2.19(2). For  $A = T_S(V)/(ab)$  in Example 2.15, the basis  $e_1, e_2, a, b, ba$  gives  $h_A(t) = 2 + 2t + t^2$ .

**Example 2.21.** For the algebras in Example 2.17, each polynomial generator may occur any number of times, whereas each exterior generator occurs at most once. Thus

$$h_{\text{Sym}(V)}(t) = \frac{1}{(1-t)^n}, \quad h_{\bigwedge V}(t) = (1+t)^n \quad \text{where } n = \dim_{\mathbb{k}} V.$$

**Lemma 2.22.** For locally finite graded vector spaces  $V, W$  concentrated in degrees bounded below,

$$h_{V \otimes_{\mathbb{k}} W}(t) = h_V(t) h_W(t), \quad (V \otimes_{\mathbb{k}} W)_n = \bigoplus_{i+j=n} V_i \otimes_{\mathbb{k}} W_j.$$

**Proof** The coefficient of  $t^n$  on either side is  $\sum_{i+j=n} \dim_{\mathbb{k}} V_i \dim_{\mathbb{k}} W_j$ . Only finitely many summands are non-zero because both gradings are bounded below.  $\square$

## 2.6 The degree 2 relations of a quotient

Let  $S$  be finite-dimensional semisimple and  $V$  a finite-dimensional  $S$ -bimodule. Let  $A = T_S(V)/I$  with  $I_0 = I_1 = 0$  as in Definition 2.10. In degree 2,

$$A_2 \cong (V \otimes_S V)/I_2, \quad \text{where } I_2 = I \cap (V \otimes_S V).$$

**Definition 2.23.** The *quadratic part* of  $A$  is  $qA := T_S(V)/(I_2)$ .

The inclusion  $(I_2) \subset I$  induces a surjective graded homomorphism  $\pi_A : qA \twoheadrightarrow A$ , and

$$\pi_A \text{ is an isomorphism} \iff I = (I_2) \iff A \text{ is quadratic.}$$

**Example 2.24.** For  $A = \mathbb{k}A_4/(e_{14})$  from Example 2.16, we have  $I_2 = 0$ , so

$$qA = \mathbb{k}A_4, \quad \pi_A : \mathbb{k}A_4 \rightarrow \mathbb{k}A_4/(e_{14}), \quad \text{Ker } \pi_A = \mathbb{k}e_{14}.$$

Thus  $qA$  does not impose the cubic relation  $e_{12}e_{23}e_{34} = 0$ .

## 2.7 Orthogonal complements and quadratic duality

**Definition 2.25.** For an  $S$ -sub-bimodule  $U$  of a finite-dimensional  $S$ -bimodule  $M$ , define the *right orthogonal complement* and *left orthogonal complement* by

$$U^\perp := \{f \in M^* \mid f(U) = 0\}, \quad {}^\perp U := \{g \in {}^*M \mid g(U) = 0\}.$$

Both  $U^\perp$  and  ${}^\perp U$  are  $S$ -sub-bimodules by (1.4). Fix a symmetrising form  $\tau$  as in Definition 1.29. For the comparison maps  $\alpha_M(f) = \tau \circ f$  and  $\beta_M(g) = \tau \circ g$  in (1.5),

$$\alpha_M(U^\perp) = \beta_M({}^\perp U) = \{\lambda \in D(M) \mid \lambda(U) = 0\}. \quad (2.8)$$

Indeed, if  $\alpha_M(f)(U) = 0$ , then

$$\tau(f(u)s) = \tau(f(us)) = 0 \quad \text{for all } s \in S, u \in U,$$

so non-degeneracy of  $\tau$  gives  $f(U) = 0$ . For  $\beta_M$ , use  $\tau(sg(u)) = \tau(g(su))$  instead.

For  $A = T_S(V)/(R)$ , the map  $\theta_r$  in (2.1) identifies  $R^\perp$  with a sub-bimodule of  $V^* \otimes_S V^*$ . These are the generators and relations in Definition 3.1:

$$V^* = \text{Hom}_S(V, S), \quad R^\perp \subset V^* \otimes_S V^*.$$

The generators  $V^*$  are assigned degree 1. In contrast, the graded  $S$ -dual  $V^{(*)}$  from Definition 1.32 has degree  $-1$  when  $V = A_1$ .

**Example 2.26.** For  $\text{Sym}(V)$  and  $\bigwedge V$  in Example 2.17, take  $V = \mathbb{k}x \oplus \mathbb{k}y$ . The relation spaces are

$$\begin{aligned} R_{\text{Sym}(V)} &= \mathbb{k}(x \otimes y - y \otimes x), \\ R_{\bigwedge V} &= \mathbb{k}(x \otimes x) \oplus \mathbb{k}(y \otimes y) \oplus \mathbb{k}(x \otimes y + y \otimes x). \end{aligned}$$

Let  $x^*, y^* \in V^*$  be the coordinate functionals. With the tensor evaluation  $\theta_r$  in (2.2),

$$\begin{aligned} R_{\text{Sym}(V)}^\perp &= \text{span}_{\mathbb{k}}\{x^* \otimes x^*, y^* \otimes y^*, x^* \otimes y^* + y^* \otimes x^*\}, \\ R_{\bigwedge V}^\perp &= \mathbb{k}(x^* \otimes y^* - y^* \otimes x^*). \end{aligned}$$

Thus the quadratic dual presentations exchange the symmetric and exterior relations.

**Exercise 2.27.** Let  $A = \mathbb{k}A_4/(e_{13}, e_{24})$  with the grading of Example 1.7. Find a homogeneous basis, compute  $h_A(t)$ , and write  $A = T_S(V)/(R)$  with explicit  $S, V, R$ .

**Exercise 2.28.** For the bimodule  $V$  in Example 2.15, find a basis of each  $T_S(V)_d$  and show that  $h_{T_S(V)}(t) = 2/(1-t)$ . Compare with the quotient  $T_S(V)/(ab)$ .

**Exercise 2.29.** For the algebras in Example 2.17, prove

$$\dim_{\mathbb{k}} \operatorname{Sym}(V)_2 = \frac{n(n+1)}{2}, \quad \dim_{\mathbb{k}} \bigwedge^2 V = \frac{n(n-1)}{2} \quad \text{where } n = \dim_{\mathbb{k}} V.$$

### 3 Lecture 3: The quadratic dual

#### 3.1 Generators and orthogonal relations

Throughout this lecture,  $S$  is a finite-dimensional semisimple  $\mathbb{k}$ -algebra and  $A = T_S(V)/(R)$  is quadratic, with  $V$  finite-dimensional and  $R \subset V \otimes_S V$  an  $S$ -sub-bimodule. Recall that  $V^* = \text{Hom}_S(V, S)$  uses right  $S$ -linear maps, whereas  ${}^*V = \text{Hom}_{S^{\text{op}}}(V, S)$  uses left  $S$ -linear maps. The tensor evaluations in (2.2) are

$$\begin{aligned}\theta_r(g \otimes f)(v \otimes w) &= g(f(v)w), \quad f, g \in V^*, \\ \theta_\ell(g \otimes f)(v \otimes w) &= f(vg(w)), \quad f, g \in {}^*V.\end{aligned}\tag{3.1}$$

Both reverse the order of the factors. The quadratic dual replaces  $V$  by an  $S$ -dual and  $R$  by the maps vanishing on  $R$ .

**Definition 3.1.** The *right quadratic dual* and *left quadratic dual* of  $A$  are

$$\begin{aligned}A^\dagger &:= T_S(V^*)/(R^\perp), \quad R^\perp = \{F \in V^* \otimes_S V^* \mid \theta_r(F)(R) = 0\}, \\ {}^\dagger A &:= T_S({}^*V)/({}^\perp R), \quad {}^\perp R = \{F \in {}^*V \otimes_S {}^*V \mid \theta_\ell(F)(R) = 0\}.\end{aligned}\tag{3.2}$$

The generators  $V^*$  and  ${}^*V$  have degree 1.

We use  $A^\dagger$  unless a side is specified. The construction uses the ordinary  $S$ -dual of  $V$ , placed in degree 1, not the graded  $S$ -dual  $V^{(*)}$ , which has degree  $-1$ . It is independent of a chosen presentation:  $V = A_1$  and  $R = \text{Ker}(A_1 \otimes_S A_1 \rightarrow A_2)$  are intrinsic to  $A$ .

**Proposition 3.2.** The algebra  $A^\dagger$  is quadratic and

$$A_0^\dagger = S, \quad A_1^\dagger = V^*, \quad A_2^\dagger \cong R^*.$$

In particular,  $\dim_{\mathbb{k}} A_2^\dagger = \dim_{\mathbb{k}} R$ .

**Proof** Restriction to  $R$  gives an exact sequence of  $S$ -bimodules

$$0 \longrightarrow R^\perp \longrightarrow V^* \otimes_S V^* \xrightarrow{F \mapsto \theta_r(F)|_R} R^* \longrightarrow 0.\tag{3.3}$$

Surjectivity follows because  $R$  is a direct summand of  $V \otimes_S V$  as a right  $S$ -module, so a right  $S$ -linear map  $R \rightarrow S$  extends to  $V \otimes_S V$ . Now use Proposition 2.12; the dimension statement follows from Proposition 1.31.  $\square$

Adding relations has the opposite effect after dualising. For  $S$ -sub-bimodules  $R \subset R' \subset V \otimes_S V$ ,

$$R \subset R' \implies (R')^\perp \subset R^\perp \implies (T_S(V)/(R'))^\dagger \twoheadrightarrow (T_S(V)/(R))^\dagger.$$

#### 3.2 Recovering the original algebra

**Lemma 3.3.** Let  $U \subset M$  be an  $S$ -sub-bimodule of a finite-dimensional  $S$ -bimodule. Under the evaluation isomorphisms of Lemma 1.35,

$${}^\perp(U^\perp) = U, \quad ({}^\perp U)^\perp = U.$$

**Proof** Every  $u \in U$  is annihilated by every  $f \in U^\perp$ , so  $U \subset {}^\perp(U^\perp)$ . If  $m \notin U$ , then  $m + U \neq 0$  in  $M/U$ . Since  $M/U$  is a direct summand of some  $S^{\oplus n}$  as a right module, a coordinate map gives a right  $S$ -linear map  $M/U \rightarrow S$  non-zero on  $m + U$ . Its composite with  $M \rightarrow M/U$  belongs to  $U^\perp$  and does not vanish on  $m$ . Hence  $m \notin {}^\perp(U^\perp)$ . The proof of  $({}^\perp U)^\perp = U$  uses left modules instead.  $\square$

**Theorem 3.4.** *Left and right quadratic duality are canonically inverse:*

$${}^!(A^!) \cong A \cong ({}^!A)^! \quad \text{as graded algebras over } S.$$

**Proof** The evaluation isomorphism  $V \cong {}^*(V^*)$  from Lemma 1.35 identifies the generators of  $A$  and  ${}^!(A^!)$ . Under the tensor evaluation  $\theta_\ell$  in (3.1), the tensor  $v \otimes w$  then acts on  $g \otimes f \in V^* \otimes_S V^*$  by  $g(f(v)w)$ . Thus the two evaluations in (3.1) agree under biduality, and Lemma 3.3 identifies  ${}^\perp(R^\perp)$  with  $R$ . The presentations (3.2) give  ${}^!(A^!) \cong A$ . Interchanging left and right gives  $({}^!A)^! \cong A$ .  $\square$

**Proposition 3.5.** *A choice of symmetrising form  $\tau : S \rightarrow \mathbb{k}$  gives isomorphisms  $A^! \cong {}^!A$  and  $(A^!)^! \cong A$  of graded algebras over  $S$ .*

**Proof** The maps  $\alpha_V(f) = \tau \circ f$  and  $\beta_V(g) = \tau \circ g$  in (1.5) give a bimodule isomorphism  $\beta_V^{-1}\alpha_V : V^* \rightarrow {}^*V$ . For  $f, g \in V^*$ , write  $f' = \beta_V^{-1}\alpha_V(f)$  and  $g' = \beta_V^{-1}\alpha_V(g)$ . Then

$$\begin{aligned} \tau(f'(vg'(w))) &= \tau(f(v)g'(w)) = \tau(g'(f(v)w)) \\ &= \tau(g(f(v)w)) \quad (v, w \in V). \end{aligned}$$

By (2.8),  $(\beta_V^{-1}\alpha_V)^{\otimes S^2}$  therefore identifies  $R^\perp$  with  ${}^\perp R$ . It induces  $A^! \cong {}^!A$  on the presentations (3.2). Apply the same comparison to  $A^!$  and then Theorem 3.4 to obtain  $(A^!)^! \cong {}^!(A^!) \cong A$ .  $\square$

Theorem 3.4 needs no choice of form. Proposition 3.5 permits the usual statement that quadratic duality is an involution, but the identification of left and right duals uses  $\tau$ .

### 3.3 Tensor algebras and monomial relations

**Example 3.6.** *Taking  $R = 0$  or  $R = V \otimes_S V$  in (3.2) gives*

$$\begin{aligned} (T_S(V))^! &= T_S(V^*)/(V^* \otimes_S V^*) \cong S \oplus V^*, \\ (T_S(V)/(V \otimes_S V))^! &= T_S(V^*). \end{aligned}$$

*In the first algebra,  $(V^*)^2 = 0$  and  $(s, f)(t, g) = (st, sg + ft)$ . Thus a tensor algebra with no relations is dual to an algebra in which every product of two positive degree elements is zero.*

For the rest of this subsection, take  $S = \bigoplus_{i=1}^r \mathbb{k}e_i$ , with  $e_i e_j = \delta_{ij} e_i$  and  $\sum_i e_i = 1$ . Choose a basis  $\mathcal{B}_{ij}$  of each  $e_i V e_j$  and let  $\mathcal{B} = \bigcup_{i,j} \mathcal{B}_{ij}$ . A word  $b_1 \cdots b_d$  in  $\mathcal{B}$  represents a non-zero basis tensor precisely when the adjacent idempotents match, as in (2.4). We call such a word *composable*.

**Definition 3.7.** *The presentation  $A = T_S(V)/(R)$  is *monomial quadratic* with respect to  $\mathcal{B}$  if  $R$  is spanned by a subset  $\mathcal{R}$  of the composable words of length 2 in  $\mathcal{B}$ .*

For  $b \in \mathcal{B}_{ij}$ , define  $b^* \in V^*$  by

$$b^*(c) = \delta_{bc} e_j \quad (c \in \mathcal{B}), \quad e_j b^* e_i = b^*. \quad (3.4)$$

The elements  $b^*$  form a basis of  $V^*$ ; their left and right idempotents are reversed.

**Proposition 3.8.** *For a monomial quadratic presentation:*

- (1) *A basis of  $A$  consists of the  $e_i$  and the composable words containing no adjacent pair from  $\mathcal{R}$ .*
- (2) *The quadratic dual is monomial, with*

$$R^\perp = \text{span}_{\mathbb{k}}\{c^* b^* \mid bc \text{ is composable and } bc \notin \mathcal{R}\}. \quad (3.5)$$

**Proof** The ideal generated by  $\mathcal{R}$  is spanned by the composable words containing an element of  $\mathcal{R}$  as an adjacent pair. Removing those basis words proves (1). For  $b \in \mathcal{B}_{ij}$  and  $c \in \mathcal{B}_{j\ell}$ , (3.1) and (3.4) give

$$\theta_r(c^* \otimes b^*)(b \otimes c) = c^*(e_j c) = e_\ell.$$

The functional vanishes on every other basis tensor. Hence  $c^* b^*$  belongs to  $R^\perp$  exactly when  $bc \notin \mathcal{R}$ , proving (2).  $\square$

**Example 3.9.** For  $A = \mathbb{k}A_n$ , take  $S = \bigoplus_i \mathbb{k}e_{ii}$  and  $V = \bigoplus_{i=1}^{n-1} \mathbb{k}e_{i,i+1}$  as in Example 2.13. There are no quadratic relations, so  $A^\dagger$  has the reversed generators  $e_{i,i+1}^* \in e_{i+1,i+1} V^* e_{ii}$  and every composable product of two generators is zero. Consequently  $h_{A^\dagger}(t) = n + (n-1)t$ . For  $n = 3$ ,

$$(\mathbb{k}A_3)^\dagger \cong (\mathbb{k}A_3/(e_{13}))^{\text{op}}, \quad (\mathbb{k}A_3/(e_{13}))^\dagger \cong (\mathbb{k}A_3)^{\text{op}}.$$

These are graded isomorphisms fixing the diagonal idempotents.

**Example 3.10.** Take  $A = T_S(V)/(ab)$  from Example 2.15, where  $S = \mathbb{k}e_1 \oplus \mathbb{k}e_2$ ,  $e_1 a e_2 = a$ , and  $e_2 b e_1 = b$ . The only composable words of length 2 are  $ab$  and  $ba$ . Thus

$$A^\dagger = T_S(V^*)/(a^* b^*), \quad A^\dagger = S \oplus \mathbb{k}a^* \oplus \mathbb{k}b^* \oplus \mathbb{k}b^* a^*.$$

The original relation  $ab = 0$  corresponds to the surviving word  $b^* a^*$ ; the surviving word  $ba$  corresponds to the dual relation  $a^* b^* = 0$ . In both algebras the Hilbert series is  $2 + 2t + t^2$ .

### 3.4 Symmetric and exterior algebras

**Proposition 3.11.** For a finite-dimensional  $\mathbb{k}$ -vector space  $V$ ,

$$\text{Sym}(V)^\dagger \cong \bigwedge D(V), \quad (\bigwedge V)^\dagger \cong \text{Sym}(D(V)).$$

**Proof** Here  $S = \mathbb{k}$  and  $V^* = D(V)$ . Choose a basis  $x_1, \dots, x_n$  of  $V$ , with coordinate functionals  $x_1^*, \dots, x_n^*$ . The evaluation (3.1) reads

$$\theta_r(x_j^* \otimes x_i^*)(x_p \otimes x_q) = \delta_{ip} \delta_{jq}.$$

For  $F = \sum_{i,j} c_{ij} x_i^* \otimes x_j^*$ , vanishing on each  $x_p \otimes x_q - x_q \otimes x_p$  is equivalent to  $c_{pq} = c_{qp}$ . Thus the orthogonal complement of the symmetric relations is spanned by  $(x_i^*)^2$  and  $x_i^* x_j^* + x_j^* x_i^*$  for  $i < j$ , which are the exterior relations. Vanishing on the exterior relations instead gives  $c_{ii} = 0$  and  $c_{pq} + c_{qp} = 0$ . The orthogonal complement is therefore spanned by  $x_i^* x_j^* - x_j^* x_i^*$  for  $i < j$ . Now use the presentations in (2.7).  $\square$

The proof works in characteristic 2: the relations  $(x_i^*)^2 = 0$  remain part of the exterior presentation even when the signs  $+$  and  $-$  coincide. Example 2.26 is the calculation for two generators.

### 3.5 Direct sums

For graded algebras  $A, B$ , the finite direct sum  $A \oplus B$  has componentwise multiplication, unit  $(1_A, 1_B)$ , and  $(A \oplus B)_i = A_i \oplus B_i$ . In particular, its degree zero algebra is  $A_0 \oplus B_0$ .

**Proposition 3.12.** Let  $A = T_S(V)/(R)$  and  $B = T_T(W)/(Q)$  be quadratic algebras. Then  $A \oplus B$  is quadratic over  $S \oplus T$ , and

$$(A \oplus B)^\dagger \cong A^\dagger \oplus B^\dagger.$$

**Proof** Regard  $V$  and  $W$  as  $(S \oplus T)$ -bimodules supported on the two respective factors. Mixed tensors vanish, so

$$\begin{aligned} T_{S \oplus T}(V \oplus W) &\cong T_S(V) \oplus T_T(W), \\ (V \oplus W) \otimes_{S \oplus T} (V \oplus W) &\cong (V \otimes_S V) \oplus (W \otimes_T W). \end{aligned}$$

The relations are  $R \oplus Q$ . Right module homomorphisms also split over the two factors, and their annihilator is  $R^\perp \oplus Q^\perp$ . Apply Definition 3.1.  $\square$

For example,  $(\mathbb{k}A_2 \oplus \mathbb{k}A_2)^\dagger \cong (\mathbb{k}A_2)^{\text{op}} \oplus (\mathbb{k}A_2)^{\text{op}}$ . Both degree zero algebras are  $\mathbb{k}^{\oplus 4}$ ; there are no products connecting the two summands.

### 3.6 Tensor products and signs

**Definition 3.13.** The *ordinary tensor product* of graded  $\mathbb{k}$ -algebras  $A, B$  has underlying vector space  $A \otimes_{\mathbb{k}} B$ , total grading  $(A \otimes_{\mathbb{k}} B)_n = \bigoplus_{i+j=n} A_i \otimes_{\mathbb{k}} B_j$ , and multiplication

$$(a \otimes b)(a' \otimes b') = aa' \otimes bb'.$$

The *graded tensor product*  $A \otimes_{\mathbb{k}}^{\text{gr}} B$  has the same graded vector space, with multiplication

$$(a \otimes b)(a' \otimes b') = (-1)^{\deg b \deg a'} aa' \otimes bb' \quad \text{for homogeneous } a, a', b, b'. \quad (3.6)$$

In the ordinary tensor product,  $a \otimes 1$  and  $1 \otimes b$  commute. In the graded tensor product they satisfy

$$(1 \otimes b)(a \otimes 1) = (-1)^{\deg a \deg b} (a \otimes 1)(1 \otimes b).$$

The sign does not change the underlying graded vector space, so both tensor products have Hilbert series  $h_A(t)h_B(t)$  by Lemma 2.22.

**Proposition 3.14.** Let  $A = T_S(V)/(R)$  and  $B = T_T(W)/(Q)$  be quadratic algebras, and assume that  $S \otimes_{\mathbb{k}} T$  is semisimple. Then both tensor products are quadratic over  $S \otimes_{\mathbb{k}} T$ , and

$$(A \otimes_{\mathbb{k}} B)^\dagger \cong A^\dagger \otimes_{\mathbb{k}}^{\text{gr}} B^\dagger, \quad (A \otimes_{\mathbb{k}}^{\text{gr}} B)^\dagger \cong A^\dagger \otimes_{\mathbb{k}} B^\dagger. \quad (3.7)$$

The assumption on  $S \otimes_{\mathbb{k}} T$  holds, for example, when both  $S$  and  $T$  are products of matrix algebras over  $\mathbb{k}$ . Over a general field it does not follow from semisimplicity of  $S$  and  $T$  alone.

**Proof** The degree 1 part of  $A \otimes_{\mathbb{k}} B$  is  $(V \otimes_{\mathbb{k}} T) \oplus (S \otimes_{\mathbb{k}} W)$ . Its quadratic relations consist of  $R \otimes_{\mathbb{k}} T$ ,  $S \otimes_{\mathbb{k}} Q$ , and the mixed relations

$$(v \otimes 1)(1 \otimes w) - (1 \otimes w)(v \otimes 1) \quad (v \in V, w \in W). \quad (3.8)$$

Here the products are taken in the tensor algebra on the degree 1 part, and their span is an  $(S \otimes_{\mathbb{k}} T)$ -sub-bimodule. These relations allow every  $W$  letter to move past every  $V$  letter; the remaining relations are exactly the presentations of  $A$  and  $B$ . Thus they present  $A \otimes_{\mathbb{k}} B$ .

The right  $(S \otimes_{\mathbb{k}} T)$ -duals of the two generator summands are  $V^* \otimes_{\mathbb{k}} T$  and  $S \otimes_{\mathbb{k}} W^*$ , with evaluations

$$(f \otimes t)(v \otimes t') = f(v) \otimes tt', \quad (s \otimes g)(s' \otimes w) = ss' \otimes g(w).$$

These identifications hold for finite free right modules and hence for their direct summands. The pure relation spaces have orthogonal complements  $R^\perp \otimes_{\mathbb{k}} T$  and  $S \otimes_{\mathbb{k}} Q^\perp$  in the corresponding pure summands. For mixed tensors, the two reversed evaluations of  $f \in V^*$  and  $g \in W^*$  both give  $f(v) \otimes g(w)$ . Consequently the annihilator of the mixed relations (3.8) is generated by

$$(f \otimes 1)(1 \otimes g) + (1 \otimes g)(f \otimes 1) \quad (f \in V^*, g \in W^*).$$

These are the mixed relations of  $A^! \otimes_{\mathbb{k}}^{\text{gr}} B^!$ , proving the first isomorphism in (3.7). For  $A \otimes_{\mathbb{k}}^{\text{gr}} B$ , the minus sign in (3.8) is a plus sign. Its annihilator has a minus sign, giving the ordinary tensor product on the dual side.  $\square$

**Example 3.15.** Take  $A = \mathbb{k}A_2 \otimes_{\mathbb{k}} \mathbb{k}A_2$ . Its degree zero algebra has basis  $e_{ii} \otimes e_{jj}$  for  $i, j \in \{1, 2\}$ , and its degree 1 part has basis

$$a = e_{12} \otimes e_{11}, \quad b = e_{22} \otimes e_{12}, \quad c = e_{11} \otimes e_{12}, \quad d = e_{12} \otimes e_{22}.$$

The only composable words of length 2 are  $ab$  and  $cd$ , and both map to  $e_{12} \otimes e_{12}$ . Thus the relation space is  $\mathbb{k}(ab - cd)$ . Using the reversed generators from (3.4), its orthogonal complement is  $\mathbb{k}(b^*a^* + d^*c^*)$ . Hence the commuting square becomes an anticommuting square:

$$ab = cd \text{ in } A, \quad b^*a^* = -d^*c^* \text{ in } A^!, \quad h_A(t) = h_{A^!}(t) = 4 + 4t + t^2.$$

The signs agree with Proposition 3.14; in characteristic 2, the same presentations remain valid with  $-1 = 1$ .

### 3.7 Exercises

**Exercise 3.16.** For  $A = \mathbb{k}A_4/(e_{13})$ , write a presentation of  $A^!$  using reversed generators and find homogeneous bases of  $A$  and  $A^!$ . Check the double dual using Proposition 3.8.

**Exercise 3.17.** Let  $S = \mathbb{k}e_1 \oplus \mathbb{k}e_2$  and let  $V$  have basis  $a, b$  with  $e_1ae_2 = a$  and  $e_2be_1 = b$ . Compute the quadratic dual and Hilbert series of  $T_S(V)/(ab, ba)$  and of its quadratic dual.

**Exercise 3.18.** For the four generators in Example 3.15, replace the relation  $ab - cd$  by  $ab - \lambda cd$ , where  $\lambda \in \mathbb{k}$ . Compute the quadratic dual relation for every  $\lambda$ , including  $\lambda = 0$ , and recover the original relation by dualising again.

## References

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